Lesson 1

Differentiation of trigonometric functions

- **We have studied before : the differentiation of basic trigonometric functions :**

1. If $y = \sin x$, then $\frac{dy}{dx} = \cos x$

2. If $y = \cos x$, then $\frac{dy}{dx} = -\sin x$

3. If $y = \tan x$, then $\frac{dy}{dx} = \sec^2 x$

- **In this year we will recognize the differentiation of reciprocals of basic trigonometric functions.**

1. If $y = \cot x$, then $\frac{dy}{dx} = -\csc^2 x$

2. If $y = \sec x$, then $\frac{dy}{dx} = \sec x \tan x$

3. If $y = \csc x$, then $\frac{dy}{dx} = -\csc x \cot x$

Proof:

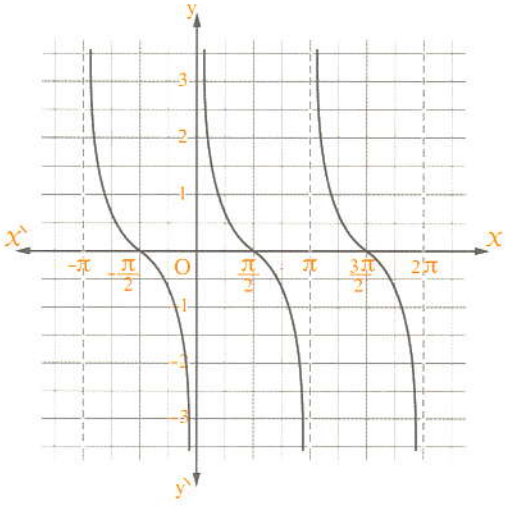
1. Cotangent function : $y = \cot x$

Where : $x \in \mathbb{R}$, $x \neq n\pi$, $n \in \mathbb{Z}$

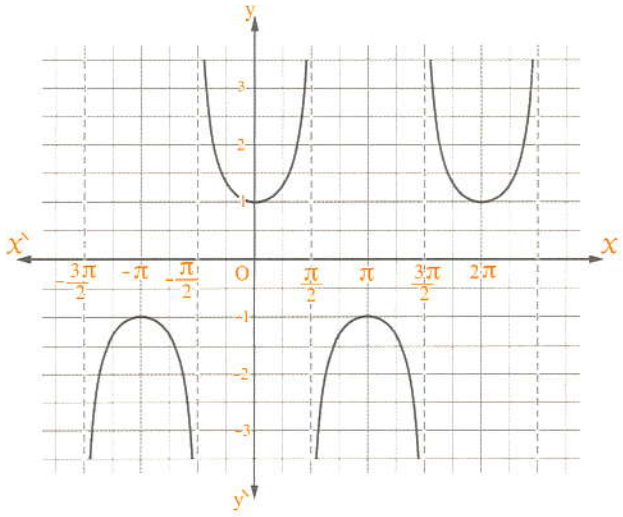
$\because y = \cot x = \frac{\cos x}{\sin x}$

$\therefore \frac{dy}{dx} = \frac{\sin x \times (-\sin x) - \cos x \times \cos x}{\sin^2 x}$

$= -\frac{1}{\sin^2 x} = -\csc^2 x$



2. Secant function : $y = \sec x$



Where $x \in \mathbb{R}$

, $x \neq \frac{(2n+1)\pi}{2}$, $n \in \mathbb{Z}$

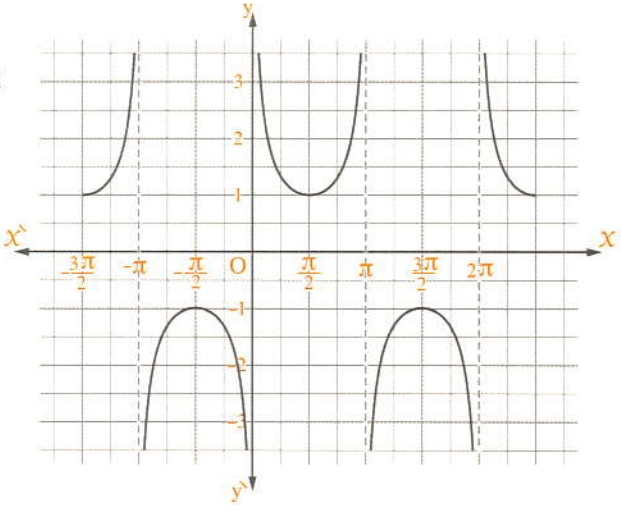
$\because y = \sec x = \frac{1}{\cos x}$

$\therefore \frac{dy}{dx} = \frac{\cos x \times \text{zero} - 1 \times (-\sin x)}{\cos^2 x}$

$= \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x} \times \frac{\sin x}{\cos x}$

$= \sec x \tan x$

3. Cosecant function : $y = \csc x$



Where : $x \in \mathbb{R}$, $x \neq n\pi$, $n \in \mathbb{Z}$

$\because y = \csc x = \frac{1}{\sin x}$

$\therefore \frac{dy}{dx} = \frac{\sin x \times \text{zero} - 1 \times \cos x}{\sin^2 x}$

$= \frac{-\cos x}{\sin^2 x} = \frac{-1}{\sin x} \times \frac{\cos x}{\sin x}$

$= -\csc x \cot x$

Generally

1. If $y = \cot z$, then $\frac{dy}{dx} = (-\csc^2 z)\frac{dz}{dx}$
2. If $y = \sec z$, then $\frac{dy}{dx} = (\sec z \tan z)\frac{dz}{dx}$
3. If $y = \csc z$, then $\frac{dy}{dx} = (-\csc z \cot z)\frac{dz}{dx}$

Where z is a differentiable function with respect to the variable x

Example 1

Find the first derivative for each of the following :

1. $y = \cot(2x - 1)$

2. $y = \sec(3x + 5)$

3. $y = 4\tan x - \csc 3x$

4. $y = x^4 + \cot(5x^3 - 2)$

5. $y = 5x^2 + \csc\sqrt{x-2}$

6. $y = \sec\frac{x}{3} + \sec\frac{\pi}{3}$

Solution

1. $\frac{dy}{dx} = -\csc^2(2x - 1) \times 2 = -2\csc^2(2x - 1)$

2. $\frac{dy}{dx} = \sec(3x + 5)\tan(3x + 5) \times 3 = 3\sec(3x + 5)\tan(3x + 5)$

3. $\frac{dy}{dx} = 4\sec^2 x - (-\csc 3x \cot 3x) \times 3$

$= 4\sec^2 x + 3\csc 3x \cot 3x$

4. $\frac{dy}{dx} = 4x^3 - \csc^2(5x^3 - 2) \times 15x^2$

$= 4x^3 - 15x^2\csc^2(5x^3 - 2)$

5. $\frac{dy}{dx} = 10x - \csc\sqrt{x-2}\cot\sqrt{x-2} \times \frac{1}{2\sqrt{x-2}}$

$= 10x - \frac{\csc\sqrt{x-2}\cot\sqrt{x-2}}{2\sqrt{x-2}}$

6. $\frac{dy}{dx} = \frac{1}{3}\sec\frac{x}{3}\tan\frac{x}{3} + \text{zero} = \frac{1}{3}\sec\frac{x}{3}\tan\frac{x}{3}$

Remark

$\because \sec\frac{\pi}{3} = 2$ (is constant value)

$\therefore$ Its derivative = zero

Example 2

Find the first derivative for each of the following :

1. $y = \cot 6x^3 + \cot^3 6x$

2. $y = x^4\cot 5x$

3. $y = (5 - 3\csc x)^4$

4. $y = \sin x \sec 3x$

5. $y = \frac{1 - \cot x}{1 + \cot x}$

6. $y = \sin(\csc 5x^3)$

Solution

1. $\frac{dy}{dx} = -\csc^2 6x^3 \times (18x^2) + 3\cot^2 6x \times -\csc^2 6x \times 6$

$= -18x^2 \csc^2 6x^3 - 18\cot^2 6x \csc^2 6x$

2. $\frac{dy}{dx} = x^4 \times -\csc^2 5x \times 5 + \cot 5x \times 4x^3$

$= -5x^4 \csc^2 5x + 4x^3 \cot 5x$

3. $\frac{dy}{dx} = 4(5 - 3\csc x)^3 \times (3\csc x \cot x)$

$= 12 \csc x \cot x (5 - 3\csc x)^3$

4. $\frac{dy}{dx} = \sin x \times \sec 3x \tan 3x \times 3 + \sec 3x \cos x$

$= 3 \sin x \sec 3x \tan 3x + \sec 3x \cos x$

5. $\frac{dy}{dx} = \frac{(1 + \cot x) \times (\csc^2 x) - (1 - \cot x)(-\csc^2 x)}{(1 + \cot x)^2}$

$= \frac{\csc^2 x (1 + \cot x + 1 - \cot x)}{(1 + \cot x)^2} = \frac{2\csc^2 x}{(1 + \cot x)^2}$

6. $\frac{dy}{dx} = \cos(\csc 5x^3) \times -\csc 5x^3 \cot 5x^3 \times 15x^2$

$= -15x^2 \csc 5x^3 \cot 5x^3 \cos(\csc 5x^3)$

Example 3

If $y = \sqrt{14 - 5z}$, $z = \cot 2x$, prove that : $3\frac{dy}{dx} - 10 = 0$ at $x = \frac{\pi}{8}$

Solution

$\because \frac{dy}{dz} = \frac{-5}{2\sqrt{14 - 5z}}$, $\frac{dz}{dx} = -2\csc^2 2x$

$\therefore \frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx} = \frac{-5}{2\sqrt{14 - 5z}} \times -2\csc^2 2x$

$= \frac{5\csc^2 2x}{\sqrt{14 - 5z}}$ (by substitution) $= \frac{5\csc^2 2x}{\sqrt{14 - 5\cot 2x}}$

$\therefore \left[\frac{dy}{dx}\right]_{x = \frac{\pi}{8}} = \frac{5\csc^2 \frac{\pi}{4}}{\sqrt{14 - 5\cot \frac{\pi}{4}}} = \frac{10}{3}$ $\quad\quad$ $\therefore 3\frac{dy}{dx} - 10 = 3 \times \frac{10}{3} - 10 = 0$

Example 4

Find the slope of the tangent to the curve of the function f where $f(X) = \sec X + \tan X$ at the point $(0 , 1)$

Solution

$\because f(X) = \sec X + \tan X$

$\therefore \grave{f}(X) = \sec X \tan X + \sec^2 X$

At the point $(0 , 1)$ $\quad \therefore \grave{f}(0) = \sec(0) \tan(0) + \sec^2(0) = 1$

$\therefore$ The slope of the tangent to the function $= \grave{f}(0) = 1$

Example 5

If $y = \csc^n X$, prove that : $\frac{dy}{dX} + n\, y \cot X = 0$

Solution

$\because y = \csc^n X = (\csc X)^n$

$\therefore \frac{dy}{dX} = n(\csc X)^{n-1} \times (-\csc X \cot X)$

$= -n(\csc X)^n \cot X$

$\therefore \frac{dy}{dX} + n \csc^n X \cot X = 0$ $\quad \therefore \frac{dy}{dX} + n\, y \cot X = 0$